MILITARY SPECIFICATION

ELBOWS; ELECTRICAL CONDUIT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for Electrical Conduit Elbows.

1.2 Classification. Elbows shall be furnished in the design specified (See 3.2).

1.2.1 Part Number. The part number shall consist of the letter "M" followed by the basic specification number and sequentially assigned dash number which is used to designate the Elbow configuration.

Example:

M 6719-1

Basic Specification Number

Dash Number (See Figure 1)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating; Cadmium (Electrodeposited)
WW-T-791	Tubing; Brass, Seamless
QQ-B-637	Brass, Naval, Rod, Wire, Shapes Forgings and Flat Products with Finished Edges (Bar, Flatwire, and Strap)
PPP-B-566	Box, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-676	Box, Setup
PPP-T-60	Tape Pressure - Sensitive Adhesive, Waterproof, for Packaging
PPP-T-76	Tape Pressure - Sensitive, Adhesive Paper, Water Resistant (For Carton Sealing)

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Military

MIL-P-116	Preservation, Methods of
MIL-C-3702	Cable, Power, Electrical Ignition, High Tension
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials used in the manufacture of shielded elbows shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications.

3.1.1 Metals. Brass conforming to Specification QQ-B-637, halfhard, may be used for the nut and male thread. Brass conforming to Specification WW-T-791, grade 1, type D, full annealed, may be used for the tube.

3.2 Design and Construction.

3.2.1 Dimensions. The dimensions shall conform to Figure 1.

3.2.2 Weight. The weight shall not exceed .075 Pound.

3.2.3 Method of Fabrication. The method of fabrication is optional with the manufacturer.

3.2.4 Finish. The elbows shall be suitably corrosion resistant. Brass as specified in paragraph 3.1.1 need not be treated to resist corrosion. Steel elbows shall be cadmium plated in accordance with Specification QQ-P-416, except that the cadmium plating shall have a minimum thickness of .0005 inch.

3.2.5 Fit of Nut. The nut shall swivel freely about the elbow and shall freely move up on the elbow until the bottom edge of the nut is at least flush with the bottom edge of the elbow.

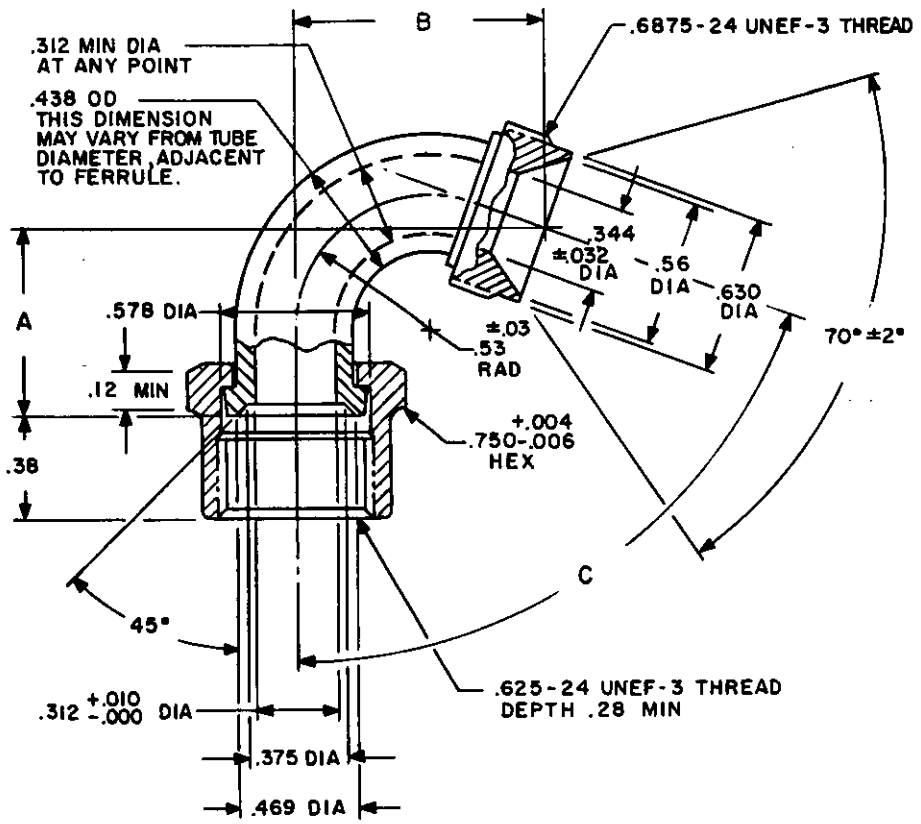
3.2.6 Threads. Threads shall be in accordance with Specification MIL-S-7742.

3.2.7 Internal Clearance. The internal passage shall be free from burrs and sharp edges and shall allow easy insertion of 7 mm cable conforming to Specification MIL-C-3702.

3.3 Performance.

3.3.1 Torque. The elbow, and nut, shall not twist or otherwise deform and the threads shall not be damaged when subjected to the Torque test described in paragraph 4.4.1.

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PART NO.	A ±.06	B ±.03	C ±5°
M6719-1	.75	1.00	70°
-2	1.06	.97	110°
-3	.89	.91	90°

NOTES:
1. REMOVE ALL BURRS AND SHARP EDGES.
2. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: TWO PLACE DECIMALS ±.02, THREE PLACE DECIMALS ±.010, ANGLES ±5°.

FIGURE 1 Outline dimensions electrical conduit elbow

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3.4 Marking. The Military Part No. shall be permanently and legibly marked on the elbow, in accordance with the general marking requirements of MIL-STD-130.

3.5 Workmanship. Workmanship shall be in accordance with high-grade aircraft practice covering this class of work.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. Inspection shall be classified as follows:

Quality conformance inspection (See 4.3).

Inspection of preparation for delivery (See 4.5).

4.3 Quality Conformance Inspection

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Visual Examination. Samples selected in accordance with 4.3.1 shall be visually examined for defects in material, design, construction, marking and workmanship. Any departure from specified requirements shall be considered a defect. The inspection level shall be level II with an AQL of 4.0 expressed as defects per hundred units.

4.3.3 Dimensional Examination. Samples selected in accordance with 4.3.1 shall be examined to determine compliance with specified dimensional requirements. Any dimension that is not within specified tolerances shall be considered a defect. The inspection level shall be level S-2 with an AQL of 6.5 expressed as defects per hundred units.

4.3.4 Testing of the End Item. Testing shall be performed on each sample unit in accordance with 4.4.1.1, 4.4.1.2 and 4.4.1.3. The inspection level shall be level S-2 and any failure to meet the specified requirements shall be cause for rejection of the entire lot.

4.4 Tests

4.4.1 Torque

4.4.1.1 Twisting. The elbow shall be attached to a fixture by the nut and a 250 pound-inch torque shall be applied to the male thread at the elbow outlet in a plane perpendicular to the axis of the outlet. A one-inch hexagonal steel nut blind tapped 11/16-24 may be used for transmitting the torque to the male thread. The torque shall be applied in any direction in the designated plane at the option of the inspector. The body of the elbow shall not twist or otherwise deform and the threads shall not be damaged.

4.4.1.2 Nut Strength. The elbow shall be attached to a standard spark plug or suitable duplicate fixture by the attaching nut and 250 pound-inches torque shall be applied to the coupling nut. There shall be no evidence of binding of the coupling nut or deformation of the retaining ferrule shoulder upon application of the torque through a plane perpendicular to the outlet axis of the elbow.

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4.4.1.3 Bending. The elbow shall be attached to a fixture by the coupling nut and a bending torque applied in the plane of the elbow through a lever attached to the elbow at the male thread end. The lever arm is measured from the center of the top of the spark plug. The elbow shall be capable of withstanding 250 pound-inches of torque in either direction without deformation to the body of the elbow.

4.5 Preparation for Delivery Examination. An examination shall be made to determine compliance with the packaging, packing and marking requirements contained in Section 5. The sample unit shall be one shipping container and the lot shall be the number of containers offered at one time. Any area of nonconformance with specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed as defects per hundred units.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Preservation and packaging shall be level A or C, as specified (See 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. Elbows shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Elbows shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation Application. Preservative shall not be used.

5.1.1.4 Unit Packaging. Elbows shall be individually packaged in accordance with MIL-P-116, submethod 1C-3 insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein.

5.1.1.5 Intermediate Packaging. Elbows, packaged as described in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape, and quantities, shall be of minimum tare and cube and shall contain multiples of 5 unit packages, not to exceed 50 unit packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 unit packages.

5.1.2 Level C. Clean and dry elbows shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The packaged elbows shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, 2 inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged elbows shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closure shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged elbows shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

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5.4 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock number items to the greatest extent practicable.

6. NOTES

6.1 Intended Use. The elbows covered by this specification are intended for use in ignition shielding assemblies of aircraft engines.

6.2 Ordering Data. Requisitions, contracts, and orders should state the part number of the elbow desired, Level A or C Packaging and Level A, B or C Packing.

6.3 Cross Reference of Part Numbers.

TABLE I Cross Reference of Part Numbers

Part Number	Supersedes
M6719-1	AN3104-1
M6719-2	AN3104-2
M6719-3	AN3104-3

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Custodian:
Air Force - 80
Navy - AS

Review Activity:
Air Force - 82

User Activity:
Army - EL

Preparing Activity:
Air Force - 80

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